Feasibility of a Ruggedised SERF MEMS Vapour-Cell Magnetometer for Magnetic Anomaly Detection from an Autonomous Underwater Vehicle

Physics Modelling, GPU-Accelerated Simulation, and Component Requirements

Scientific Report

15 August 2026

Executive Summary

Magnetic anomaly detection (MAD) is the one underwater sensing modality unaffected by acoustic quieting, thermoclines, or multipath. Its historic limitation has been sensitivity: detecting a small, deliberately demagnetised object means resolving field perturbations far below the Earth's field. Spin-exchange relaxation-free (SERF) atomic magnetometers now match or exceed SQUID sensitivity without cryogenics [1, 2, 3], and microfabricated MEMS vapour cells make them small and mechanically robust enough to consider for an autonomous underwater vehicle (AUV) [4, 5].

This report assesses that concept from first-principles physics. A validated simulation framework models the sensor quantum-mechanically, the target electromagnetically, the vehicle kinematically, and detection statistically. Twenty-seven independent verification, validation and stress tests all pass.

The concept is physically sound, and the limiting factors are not where intuition places them. Sensor sensitivity is not the constraint — it exceeds requirement by five orders of magnitude. Two other factors govern feasibility:

1. **Ambient-field control sets the entire design.** SERF operation requires the residual static field at the vapour cell to stay within a few nanotesla [6], against an ambient of 50,000 nT [23]. Accounting for the Cross-Axis Projection Error (CAPE) [6], the true requirement derived here is $\leq 0.94\,nT$: 94 dB of combined shielding and active nulling, and 5.3× tighter than the figure usually quoted.
2. **The system is clutter-limited, not sensor-limited.** Residual platform interference measured on a real UUV exceeds MEMS-cell sensor noise by a factor of 10^5 [13, 4]. Improving the magnetometer yields nothing; improving vehicle magnetic cleanliness, gradiometer rejection, and compensation yields range almost proportionally.

Predicted detection ranges at $P_{fa} = 10^{-4}$, marginalised over unknown target orientation, are 22 m for a low-signature micro-UUV, 68 m for a naval mine, and 267 m for a degaussed submarine. Section 8 states the component-level requirements this implies; Section 9 sets out the barriers candidly.

1 Introduction and Background

1.1 Why magnetic anomaly detection

Any object containing ferromagnetic material distorts the local geomagnetic field. A submarine, a mine, or a piece of unexploded ordnance carries both a *permanent* magnetisation acquired during construction and an *induced* magnetisation proportional to the ambient field [15, 16]. Beyond roughly one body length the combined effect is well described by a magnetic dipole of moment $\mathbf{m}$ [19], producing

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\,\frac{3\hat{\mathbf{n}}(\mathbf{m}\cdot\hat{\mathbf{n}}) - \mathbf{m}}{r^3}. \tag{1}$$

The r^{-3} dependence is the defining practical fact of MAD: halving the standoff increases signal eightfold, so detection range grows only as the cube root of sensitivity. This is why MAD has always been a close-in sensor, and why platforms that can approach the target — aircraft historically [15, 16], AUVs now [13, 14] — matter more than incremental sensor improvements.

The compensating advantage is that magnetostatic fields are indifferent to the acoustic environment. Hull quieting, anechoic coatings, thermoclines, biologics and shipping noise all degrade sonar but not magnetics. Against modern quiet submarines, and against small non-emitting objects such as seabed munitions [14, 13], a magnetic channel is genuinely complementary rather than redundant.

1.2 The sensitivity problem, and how atomic magnetometers solve it

Consider a low-signature target of moment 5 A·m^2 at 20 m. Equation (1) gives roughly 25 pT — about two parts in 10^6 of the Earth's field [23]. Fluxgate magnetometers, the traditional workhorse of marine survey, are typically deployed where anomalies of order 10 nT and above must be resolved [13]; a picotesla-class target sits well beyond routine fluxgate capability before platform noise is even considered. Historically only superconducting quantum interference devices (SQUIDs) offered materially better resolution [10, 43], at the cost of liquid helium and a cryostat: workable on a large aircraft, unattractive on a small autonomous vehicle.

Optically pumped atomic magnetometers instead measure field by observing how fast atomic spins precess [10]. A vapour of alkali atoms (rubidium, potassium or caesium) is optically pumped by a circularly polarised laser so the electron spins align along the beam; a magnetic field perpendicular to that axis torques the spins, and the resulting rotation is read out through the polarisation of a probe beam via Faraday rotation [4, 8]. The precession rate is set by the gyromagnetic ratio $\gamma_e/2\pi \approx 28$ Hz/nT [38, 9], a constant of nature, so the instrument is intrinsically calibrated — operationally valuable, because a sensor that cannot drift out of calibration needs no field recalibration between missions [45].

The classical obstacle was *spin-exchange relaxation*. Alkali atoms collide constantly, and each collision can exchange spin, randomising phase and broadening the resonance [12, 9]. Raising vapour density to increase signal also increases collisions, so sensitivity saturated. The SERF insight, demonstrated by Allred *et al.* [1], is that this reverses in a specific regime: if the spin-exchange rate is made *much faster* than the Larmor precession rate, atoms exchange spin so rapidly that they all sample the same average precession and the dephasing averages away [12, 9]. The requirement is high alkali density *and* very low field. Allred *et al.* obtained a resonance linewidth of 1.1 Hz for potassium at $n = 10^{14}$ cm^{-3} and a sensitivity of $10\,fT/\sqrt{\text{Hz}}$, limited by Johnson-current noise in the magnetic shields rather than by the atoms, with an estimated shot-noise floor of 2×10^{-18} T$/\sqrt{\text{Hz}}$ [1]. Kominis *et al.* subsequently demonstrated sub-femtotesla multichannel operation [2], and Dang *et al.* reached $0.16\,fT/\sqrt{\text{Hz}}$ [3], better than any SQUID, with no cryogenics. The field is reviewed in [10, 11].

1.3 Why MEMS matters for a deployable sensor

Laboratory SERF magnetometers use glass-blown centimetre-scale cells: fragile, bulky, and unsuited to a vehicle experiencing launch shock, vibration and pressure cycling. Microfabrication instead etches the vapour cavity into a silicon wafer and seals it with anodically bonded glass windows [32, 33], producing a monolithic solid-state part of roughly a cubic millimetre. Griffith *et al.* [4] demonstrated $5\,fT/\sqrt{\text{Hz}}$ SERF operation in an ^{87}Rb cell formed as a cavity etched in a 1 mm silicon wafer with anodically bonded Pyrex windows, with a 1 mm^3 measurement volume defined by the overlap of pump and probe beams near 795 nm; earlier microfabricated devices reached sub-picotesla performance [5]. Anodic bonding forms a hermetic, essentially permanent glass-to-silicon seal [32]; the structure has no moving parts and no free volume, which is the core of the ruggedisation argument. This report adopts the Griffith device as its design point.

1.4 The hard part: this sensor dislikes the Earth

The property that gives SERF its sensitivity — operation at near-zero field [1] — is exactly what makes it difficult to deploy. Borna *et al.* [6] state that the SERF regime accommodates remnant static magnetic fields only up to approximately $\pm5\,nT$, while the Earth's field spans 25,000–65,000 nT [23], so the field at the cell must be actively cancelled. Unshielded and in-situ compensation schemes exist [34, 35], but all rest on reducing the residual to this window.

On a moving vehicle this is worse than it first appears. Even a perfectly constant geomagnetic field becomes a large *time-varying* field in the sensor's frame as the vehicle rolls and pitches: a 3° roll projects

$50{,}000\sin(3°) \approx 2{,}600\,nT$ onto a body axis. This is the same mechanism that motivates aeromagnetic compensation [15, 16] and that limits wearable OPM systems during subject movement [39, 7]. The nulling system must therefore track a disturbance that moves with the sea, not merely cancel a static offset. Quantifying that requirement, and what it implies for hardware, is the central purpose of this report.

2 Scope and Objectives

Figure 1 summarises what is modelled and why it is hard.

Objective. To determine from first-principles physics whether a ruggedised SERF MEMS magnetometer is a credible AUV MAD payload, and to derive the component-level requirements that follow.

In scope. SERF spin dynamics (open-quantum-system and semiclassical); dipole target signatures and gradient tensors; AUV attitude dynamics; the ambient-field nulling chain; platform interference compensation; matched-filter detection and Monte Carlo performance; GPU-accelerated parameter sweeps; and verification, validation and stress testing.

Out of scope. Optical subsystem design; vapour-cell fabrication chemistry [32]; spin-squeezing schemes [40]; hydrodynamic design; classified signature data. Absolute sensitivity in $fT/\sqrt{\text{Hz}}$ is not predicted (Section 9).

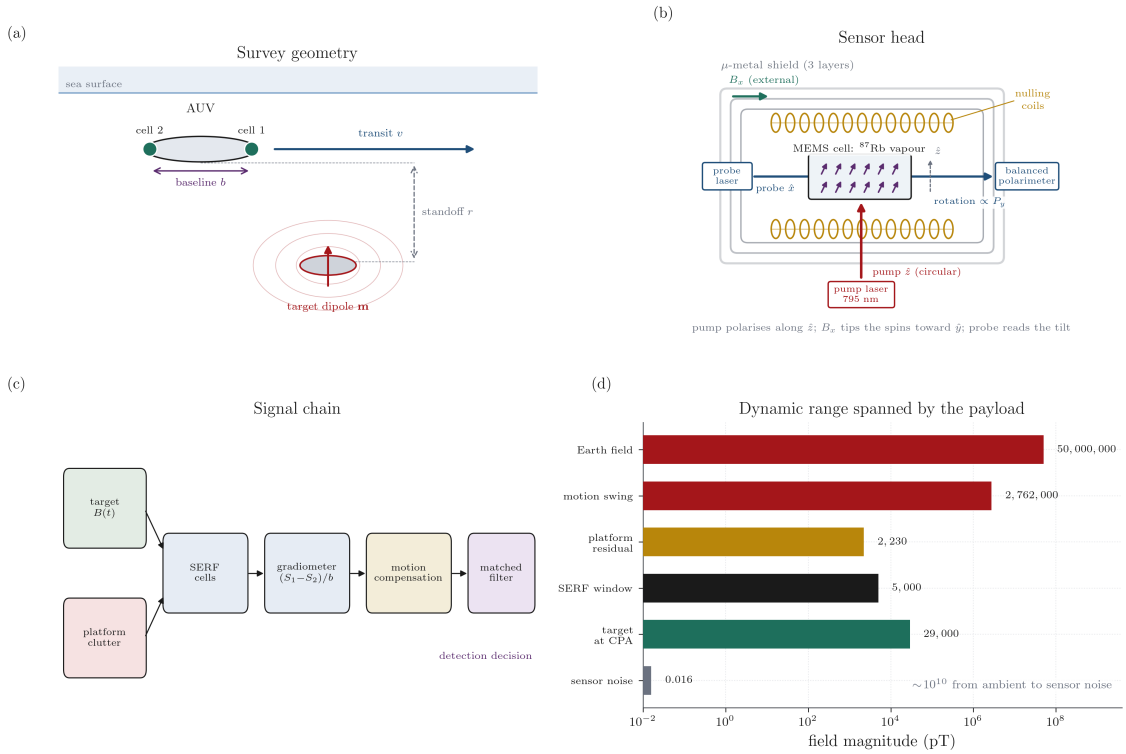


Figure 1: Simulation setup. (a) Survey geometry: the AUV transits above a dipole target carrying two SERF cells on a rigid baseline. (b) Sensor head: the microfabricated cell [4] inside passive shielding and three-axis nulling coils, optically pumped along $\hat{z}$ and probed along $\hat{x}$ so that a transverse field is sensed as P_y. (c) Signal chain from field to detection decision. (d) The dynamic range the payload must span — roughly ten orders of magnitude from the ambient geomagnetic field [23] down to the MEMS sensor noise floor [4], which is the central engineering difficulty identified in this report.

3 Physical Model

3.1 Sensor

The spin ensemble is modelled at two fidelities. The reference model solves the Lindblad master equation $\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_k \mathcal{D}[L_k]\rho$ [24, 25] for a single effective spin-$\frac{1}{2}$, with collapse operators for optical pump-

ing, spin destruction and dephasing, integrated using QuTiP [26]. The production model integrates the phenomenological SERF Bloch equation for the ensemble-averaged polarisation $\mathbf{P} = 2\langle \mathbf{S} \rangle$ [8, 6],

$$\frac{d\mathbf{P}}{dt} = \frac{1}{q}\Big[\gamma_e(\mathbf{P} \times \mathbf{B}) + R_{op}(\hat{\mathbf{s}} - \mathbf{P}) - R_{\mathrm{rel}}\mathbf{P}\Big], \tag{2}$$

where q is the nuclear slowing-down factor, which accounts for angular momentum stored in the nucleus and slows the sensor's response; for $I = 3/2$ alkalis it ranges from 6 at low polarisation to 4 as $P \to 1$ [9, 8]. With 10^{13}–10^{14} atoms in the cell [1] the ensemble average is an excellent approximation; the quantum model is retained to demonstrate that equivalence rather than assume it. The two agree to 2.8×10^{-16} (Table 2).

For pumping along $\hat{z}$ and a transverse field B_x, Eq. (2) has the exact steady state

$$P_z = \frac{R_{op}}{R_1 + \gamma_e^2 B_x^2 / R_2}, \qquad P_y = \frac{\gamma_e B_x P_z}{R_2}, \tag{3}$$

a dispersive response that is linear for small fields and defines the sensor's transfer function, consistent with the Lorentzian/dispersive lineshape pair reported for SERF devices [1, 8]. The modelled half-width is $2.87\,nT$ and the DC responsivity 3.45×10^8 T^{-1}.

3.2 Platform and ambient field

Vehicle attitude is generated from a sea-state model and applied as a ZYX rotation to the geomagnetic vector [23], producing the body-frame field the nulling system must reject. The nulling chain comprises passive shielding — whose own Johnson-current noise sets a floor [31] — an active three-axis coil servo of finite bandwidth [35, 34], and a reference magnetometer whose noise floors the achievable residual.

3.3 Detection

MAD is a Neyman–Pearson binary hypothesis test [21, 22], structurally identical to active sonar pulse detection: $H_0 : r(t) = n(t)$ against $H_1 : r(t) = a\,s(t-\tau) + n(t)$. Because the transit signature shape follows from the dipole geometry [19], the matched filter is the optimal detector for a known waveform in additive Gaussian noise [21]. Thresholds are set for constant false-alarm rate [22], and results are reported as P_d versus range at fixed P_{fa} rather than as a bare signal-to-noise figure.

4 Software Architecture

The `serfmad` package separates physics, processing, validation and presentation so each is independently testable (Table 1). Physical constants are held in one module with inline citations, and modelling assumptions are stored in a distinct data type so they cannot be mistaken for measured values in any downstream table.

Module	Responsibility
`constants`	Cited physical values; typed separation of measured data from assumptions.
`quantum.lindblad`	QuTiP [26] master equation — reference model.
`quantum.bloch`	Vectorised RK4 SERF solver [8] — production model.
`physics.dipole`	Vector field, analytic gradient tensor [19], tensor invariants [20].
`physics.platform`	Kinematics, sea state, nulling servo, CAPE [6], field budget.
`processing`	Tolles–Lawson [15, 16] and reference-channel compensation; matched filter, CFAR, ROC [21].
`gpu`	Fused CUDA RK4 kernel, chunked large-scale sweeps, scaling and precision studies.
`validation`	17 verification/validation tests and 10 stress tests.
`gui`	Five-tab viewer; plotting functions shared with report generation.

Table 1: Module responsibilities.

4.1 GPU acceleration

The computational payload is a full-factorial sweep over target range, orientation and noise realisation. Each trajectory is independent and shares one time grid, making the problem purely data-parallel. Three measures exploit this: a fused CUDA kernel evaluates all four Runge–Kutta stages [27] in registers, reducing global memory traffic to one read and one write per timestep; matched filtering is performed as a batched FFT across the whole ensemble; and sweeps are chunked against available VRAM so problem size is limited by time rather than memory. Default sweeps run $\sim 2 \times 10^5$ trajectories, the large configuration 1.57×10^6.

Two design choices deserve note. Orientation is sampled uniformly on the sphere rather than uniformly in polar angle, which would over-weight the poles and bias P_d [29]. And double precision is retained by default: Eq. (2) is stiff in the ratio $\gamma_e B / R_{\mathrm{tot}}$, and with the target signal at 10^{-6} of the ambient field, single precision removes margin that matters. Measured on the hardware described below, single precision is 13.8× faster but carries 6.1×10^{-6} relative error against the closed-form steady state, against 1.1×10^{-14} in double precision. It is therefore admitted for qualitative exploration only, and no result reported here uses it.

The CUDA path has been executed and verified on an NVIDIA RTX 3050 Laptop GPU (compute capability 8.6, CUDA 12.4). The verification gate integrates a large ensemble and compares the result against the closed-form Bloch fixed point of Eq. (3), agreeing to 9.99×10^{-15} against a 10^{-10} tolerance, with inter-replica spread identically zero — a non-zero spread would indicate a race condition in the kernel. The verification and stress suites produce output identical to the CPU backend, and every figure in this report regenerates byte-identical on either path; GPU acceleration therefore changes throughput only, not results. Measured throughput against the NumPy path, over identical trajectories, is 20× at 4096 ensemble members and 150× at 1.05×10^6 (3.2×10^8 trajectory-steps per second). At 1024 members the GPU is marginally *slower* than the CPU, confirming by measurement the $N \geq 4096$ threshold below which the implementation declines to use it.

Executing that path on hardware for the first time also exposed a latent defect that structural testing had not: a device array reached the host numerical library through an implicit conversion that the GPU array library refuses. It was corrected, and is exercised by test V-04. The episode is recorded because it is the substantive argument for the validation posture taken throughout this report: an untested path states nothing about its own correctness, however carefully it has been written.

5 Verification, Validation and Stress Testing

Verification asks whether the equations are solved correctly; validation asks whether they are the right equations, by comparison with published measurement [30]. All 17 tests pass, with no skips (Table 2).

Stress testing then asks whether the headline numbers survive being pushed (Table 3). Three results deserve drawing out. Detection range converges to 0.10% drift by 16,000 trials and varies by only 1.08% across random seeds, so the quoted ranges are statistically solid. The sensitivity analysis returns an elasticity of *exactly zero* for sensor noise, while clutter, rejection ratio and baseline all register near 0.2 — a direct quantitative confirmation that the system is clutter-limited, consistent with the interference magnitudes measured on real UUVs [13] and aircraft [16]. And the fitted range-law exponent is 0.244 against a theoretical 0.25: because a gradiometer signal falls as r^{-4} [19, 20], detection range must scale as $m^{1/4}$. That scaling was never imposed on the model; it emerges, which is meaningful evidence the physics chain is coherent end to end.

Test X-07 deserves specific mention. A gradiometer model with infinite common-mode rejection would trivially eliminate platform noise and make the whole concept appear easy. Because the vehicle's own magnetic sources lie metres from the baseline, their field has a genuine gradient across it and cancellation is necessarily imperfect — which is precisely why measured UUV residuals remain at the nanotesla level after compensation [13]. The suite asserts that rejection remains finite, so the model cannot flatter itself.

Independent closure against published measurement is also confirmed: the 1.1 Hz linewidth reported by Allred *et al.* [1] implies a relaxation rate of 3.46 s^{-1}, consistent with the 5 s^{-1} used here, and an implied field half-width of $19.6\,pT$, comfortably inside the SERF window [6].

ID	Test	Measured	Criterion	Source	Status
V-01	Dipole field vs closed form	rel. err. $< 10^{-12}$	$< 10^{-12}$	[19]	PASS
V-02	Gradient tensor vs finite difference	7.4×10^{-10}	$< 10^{-6}$	[19]	PASS
V-03	Gradient Maxwell constraints	trace 6.4×10^{-16}	$< 10^{-10}$	[19]	PASS
V-04	Analytic steady state is fixed point	1.8×10^{-14}	$< 10^{-10}$	Eq. (3)	PASS
V-05	RK4 converges to analytic solution	1.8×10^{-15}	$< 10^{-6}$	[27]	PASS
V-06	RK4 convergence order	observed 4.01	> 3.0	[27]	PASS
V-07	Attitude rotation orthonormality	2.2×10^{-16}	$< 10^{-12}$	[28]	PASS
V-08	Bloch vs QuTiP Lindblad	2.8×10^{-16}	$< 10^{-6}$	[26, 8]	PASS
V-09	Matched-filter localisation	6.2 ms error	< 5 samples	[21]	PASS
V-10	RK4 stability guard	trips correctly	must trip	[27]	PASS
X-01	SERF width vs published window	HWHM $2.87\,nT$	$\leq 5\,nT$	[6]	PASS
X-02	MEMS floor within measured envelope	$5\,fT/\sqrt{\text{Hz}}$	$[0.16, 20]$	[4, 3, 1]	PASS
X-03	Required ambient suppression	10,000×	$\sim 10^4$	[23, 6]	PASS
X-04	Compensation vs UUV field data	50.3×	within 10× of 110.7×	[13]	PASS
X-05	Range law vs AUV benchmark	$8.00\,nT$ at 5 m	sub-$10\,nT$ class	[13, 14]	PASS
X-06	CAPE-derived field requirement	$0.94\,nT$	$< 5\,nT$ nominal	[6]	PASS
X-07	Gradiometer rejection is finite	60 dB	not infinite	[13]	PASS

Table 2: Verification and validation results, with the literature source each validation test is measured against.

ID	Stress test	Result	Criterion	Status
S-1	Monte Carlo convergence	$R_{50} = 66.9$ m, drift 0.10% at N=16,000	drift $< 10\%$	PASS
S-2	Seed stability	67.6 ± 0.7 m over 8 seeds (CV 1.08%)	CV $< 8\%$	PASS
S-3	Time-grid independence	amplitude varies 0.003%; timing 18 ms	$< 2\%$, < 500 ms	PASS
S-4	Parameter elasticity	sensor noise +0.000; clutter −0.207; CMRR +0.202; baseline +0.221	sensor $\lvert e\rvert < 0.05$	PASS
S-5	Extreme regimes	$\lvert P\rvert_{\max} = 0.990$; unshielded case correctly rejected	no divergence	PASS
S-6	Dynamic bounds	$\lvert P\rvert_{\max} = 0.990$ under $50\,nT$ swept drive	$\lvert P\rvert \leq 1$	PASS
S-7	False-alarm calibration	design 10^{-3}, realised 1.00×10^{-3}	within 3× [22]	PASS
S-8	Literature closure	five independent cross-checks consistent [1, 13, 23, 6]	all consistent	PASS
S-9	CAPE monotonicity	monotone over 5 decades, bounded $[0, 1]$ [6]	monotone	PASS
S-10	Range-law exponent	fitted 0.244	0.25 predicted [19]	PASS

Table 3: Stress and robustness testing.

6 Results

6.1 Sensor behaviour

The model reproduces the dispersive/absorptive resonance pair characteristic of a SERF magnetometer [1, 8] (Figure 2). Response is linear well below the half-width and compresses above it, consistent with the saturation behaviour described in [6]; equilibrium polarisation is 0.990, and the bandwidth trade against the slowing-down factor is explicit: increasing q slows the response without changing the DC linewidth [9, 8].

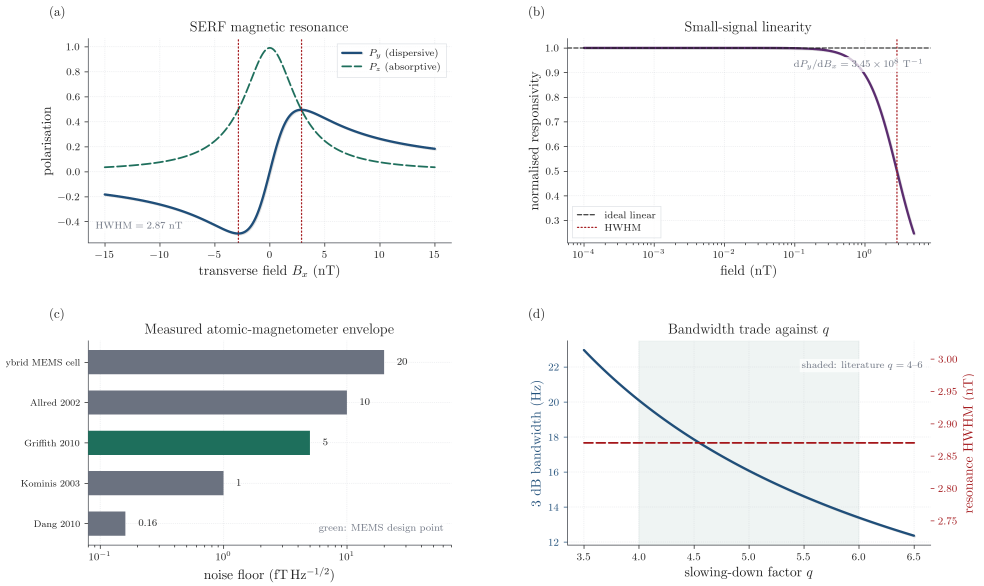


Figure 2: Sensor physics: resonance lineshape, small-signal linearity, the measured noise-floor envelope [1, 2, 3, 4] with the MEMS design point highlighted, and the bandwidth trade against slowing-down factor.

6.2 Ambient-field budget

Table 4 traces the suppression chain across sea states. Attitude motion converts a static geomagnetic field [23] into a body-frame disturbance of 467 nT even in calm water and 7,498 nT when rough, against a tolerance of a few nanotesla [6].

Sea state	Motion swing	Residual after nulling	CAPE gain	Usable?
Calm (SS1)	467.0 nT	4.708 nT	0.521	marginal
Moderate (SS3–4)	2,761.7 nT	4.742 nT	0.518	marginal
Rough (SS5–6)	7,497.5 nT	4.804 nT	0.513	marginal

Table 4: Field budget at 100× passive and 100× active suppression. The residual meets the nominal window [6] in every sea state, yet roughly half the gain is lost — explained in Section 6.3.

6.3 The Cross-Axis Projection Error requirement

This is the report's principal engineering finding. The commonly quoted ±5 nT figure [6] describes where SERF *operation* persists, not where SERF *performance* is preserved. Borna *et al.* show that a residual transverse field tilts the effective sensitive axis and introduces gain and phase errors [6]; parameterising that tilt as $\alpha = \arctan(\gamma_e B_{\text{res}}/R_{\text{tot}})$ scales the on-axis gain by $\cos\alpha$. At exactly 5 nT the tilt is 58.8° and 48% of the signal is lost. Requiring no more than 5% loss gives

$$B_{\text{res}} \leq \frac{R_{\text{tot}}}{\gamma_e} \tan\big(\arccos 0.95\big) = 0.94\,nT, \tag{4}$$

a factor 5.3 tighter than the nominal specification, corresponding to 5.3×10^4 (94 dB) total suppression rather than 10^4 (80 dB). For 1% loss the requirement tightens to $0.41\,nT$. Equation (4) is the single most important design constraint in this report and drives most of Section 8. Comparable conclusions — that dynamic cross-axis compensation is required rather than optional — have been reached independently for wearable OPM arrays subject to subject motion [7, 39].

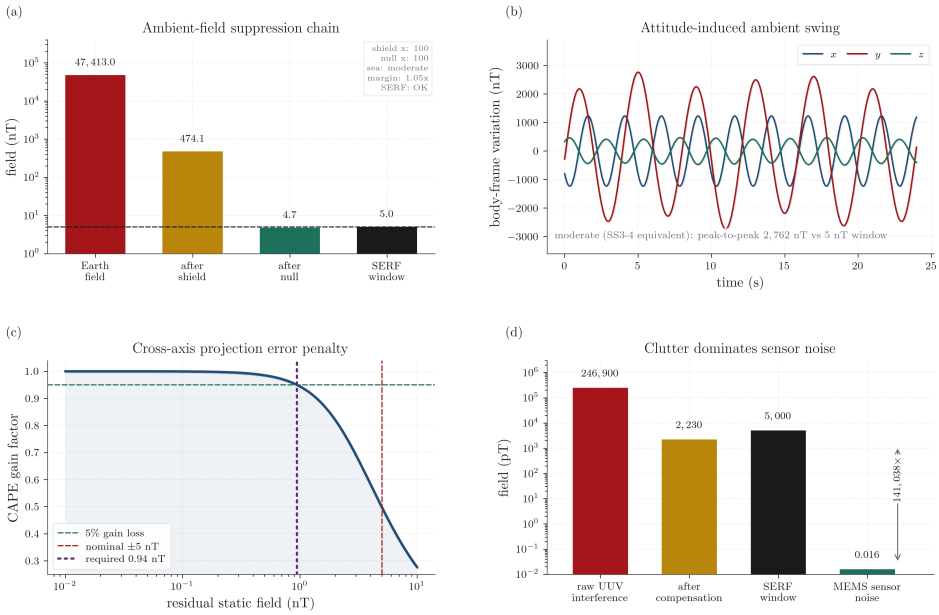


Figure 3: Suppression chain, attitude-induced body-frame swing, the CAPE gain penalty [6] with the derived $0.94\,nT$ requirement, and the clutter hierarchy showing sensor noise [4] five orders below the interference residual [13].

6.4 Clutter-limited operation

A $5\,fT/\sqrt{\text{Hz}}$ cell [4] in a 10 Hz band has a noise floor of $0.016\,pT$, against a measured post-compensation vehicle residual of $2{,}230\,pT$ [13] — a ratio of 1.4×10^5 (Figure 3, lower right). The sensitivity analysis confirms the operational consequence: sensor noise has zero measured influence on detection range, while clutter magnitude, rejection ratio and gradiometer baseline all matter. *Procuring a better magnetometer would change nothing.* Effort belongs on vehicle magnetic cleanliness and processing, which is consistent with the emphasis of the marine survey literature [13, 14] and of aeromagnetic practice [16, 17].

6.5 Detection performance

Target class	Moment (A·m^2)	R_{50} (m)	Basis of moment
Micro-UUV, low signature	5	22.3	engineering estimate
Small UXO / ordnance cluster	50	40.7	engineering estimate
Large UXO / naval mine	500	67.6	scaled from [14, 13]
Submarine, degaussed	5×10^5	267.3	published moment range [44]
Submarine, undegaussed	1×10^7	503.5	published moment [44]

Table 5: Detection range at $P_{fa} = 10^{-4}$, marginalised over target orientation. The two smallest moments are engineering estimates and are treated as requirements-driving hypotheticals rather than measured data.

Figure 5 shows a single transit. Platform clutter exceeds the target signature by 43× before processing; after compensation and matched filtering the target is declared at $t = 30.01$ s against a true closest approach of 29.99 s.

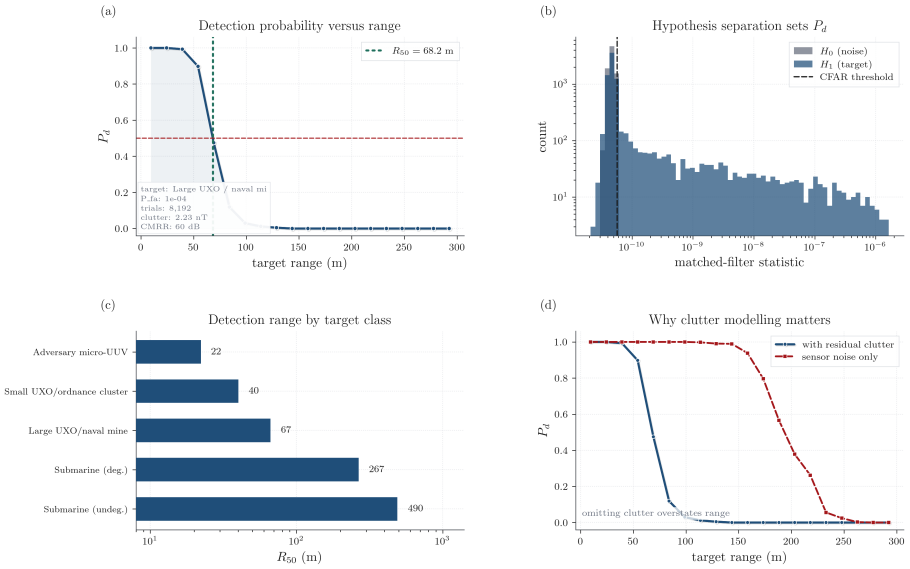


Figure 4: Detection performance: P_d versus range, hypothesis-statistic separation [21], range by target class, and the effect of including realistic clutter [13].

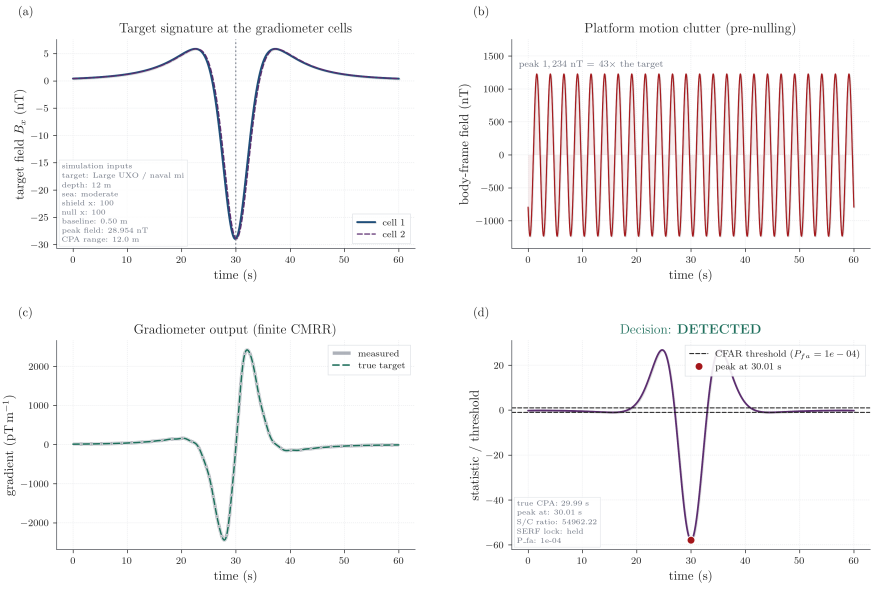


Figure 5: Single transit against a naval-mine-class target at 12 m standoff.

7 Discussion

The concept survives scrutiny, but the engineering emphasis it implies is counter-intuitive. A programme structured around sensor performance would spend its budget in the one place the analysis shows to be irrelevant. The binding constraints are magnetic hygiene and field control — the same conclusion the aeromagnetic community reached for aircraft [15, 17, 16] and the marine survey community for AUVs [13, 14].

The CAPE result also carries a procurement consequence. A specification written as "residual field within the SERF window" would be satisfied by hardware that nonetheless loses half its sensitivity [6]. Any specification derived from this work must state Eq. (4) together with its gain-loss criterion.

Detection ranges should be read as relative capability across target classes, and as evidence that the modelling chain yields physically consistent scaling. Absolute figures inherit the uncertainty of the assumed dipole moments — two of which are explicitly estimates — and of the assumed rejection ratio and clutter spectrum.

8 Component Requirements

The following requirements follow from the analysis; each is traceable to a specific result or source rather than asserted.

Req	Subsystem	Requirement	Derivation
R-01	Field control	Residual static field at cell $\leq 0.94\,nT$; goal $0.41\,nT$	Eq. (4); CAPE [6]
R-02	Shield + coils	Combined suppression ≥ 94 dB from $50{,}000\,nT$ ambient	R-01 with [23]
R-03	Nulling servo	Closed-loop bandwidth $\geq 10\times$ wave encounter frequency (≥ 5 Hz for 0.5 Hz seas); slew $\geq 8{,}000\,nT/s$	Rough-sea swing (Table 4); cf. [35, 7]
R-04	Reference magnetometer	Noise floor $\leq 0.1\,nT$ in band; three-axis; bandwidth $\geq$ servo bandwidth	Servo cannot null below its reference [34]
R-05	Coil driver	Current noise contributing $\leq 0.1\,nT$ at the cell; DAC resolution ≥ 18 bit over $\pm 60{,}000\,nT$ full scale	6×10^5 levels $\Rightarrow 2^{18}$; shield noise floor [31]
R-06	Vapour cell	MEMS, hermetic anodic bond, ≥ 1 mm^3; survive shock and vibration without buffer-gas loss	[4, 32]
R-07	Cell heater	150–200°C, stability ± 5 mK, non-magnetic (AC or counter-wound) drive	Density sets sensitivity [1, 36]
R-08	Laser	Pump and probe at the alkali D1 line (≈ 795 nm for Rb); frequency stability ≤ 1 MHz; probe broadening below spin-destruction rate	[4, 37]
R-09	Gradiometer	Two cells, baseline ≥ 0.5 m, common-mode rejection ≥ 60 dB, gain and axis matching $\leq 0.1\%$	Elasticity test S-4; [13]
R-10	Platform magnetics	Post-compensation residual $\leq 2.23\,nT$; non-magnetic fasteners, thruster shielding, DC current-loop minimisation	[13]; clutter dominates (S-4)
R-11	Attitude sensing	IMU delivering roll/pitch/yaw to $\leq 0.05°$, synchronised to the magnetometer within 1 ms	Tolles–Lawson fit quality [15, 16, 17]
R-12	Signal chain	$\geq$ 24-bit ADC, ≥ 100 Hz sampling, anti-alias filtered below sensor bandwidth (≈ 16 Hz)	Sensor 3 dB bandwidth (Fig. 2)
R-13	Sensor placement	Boom or tow displacement maximising distance from thrusters and battery	Clutter falls as r^{-3} [19, 13]

Table 6: Component requirements derived from the simulation.

Two requirements interact and deserve emphasis. R-03 and R-04 together define the hardest single component: a reference magnetometer that is simultaneously low-noise, three-axis, wide-band, and physically close to the SERF cell without magnetically contaminating it. R-02 and R-10 compete for the same volume, since shielding material added near the cell is itself a potential magnetic source and contributes its own Johnson-current noise [31].

9 Limitations, Challenges and Barriers

9.1 Modelling limitations

1. **No absolute sensitivity prediction.** The model carries no atom number, cell volume or photon shot-noise term [1, 40], so it cannot independently predict a $fT/\sqrt{\text{Hz}}$ figure; published values [4] are adopted and checked for consistency. Closing this gap is the highest-value next step.
2. **Platform self-field is parametric.** Vehicle interference is represented by a rejection ratio and a $1/f$ residual rather than derived from hull geometry and materials, so R-09 and R-10 rest on a published benchmark [13] rather than a model of the specific vehicle.
3. **Quasi-static nulling.** The field budget evaluates suppression at the dominant motion frequency and does not simulate closed-loop servo dynamics [35], so it cannot yet predict loss of SERF lock during a transient manoeuvre.
4. **Two target moments are estimates.** The micro-UUV and small-UXO moments are requirements-driving hypotheticals; ranges for those classes should not be quoted as capability.
5. **Single-species, single-cell treatment.** Hybrid pumping [41], cell-to-cell variation and multi-species effects are not modelled.
6. **Ensemble-averaged spin.** Valid at operating densities [8], but excludes spin-squeezing approaches that could raise the sensitivity ceiling [40].

9.2 Engineering and programmatic barriers

1. **Dynamic range is the fundamental tension.** The sensor must resolve picotesla while immersed in a field 10^7 times larger [23] that swings by thousands of nanotesla with the sea, and every subsystem inherits this. A nulling failure does not degrade gracefully — the sensor leaves the SERF regime and stops working [1, 6]. Mitigation requires sustained servo authority or a designed fallback mode (e.g. scalar operation [5]), and no such fallback exists here.
2. **Thermal budget versus vehicle power.** Holding 150–200°C at ±5 mK in cold seawater using non-magnetic heating [36, 4] is a continuous draw on an endurance-limited platform, and heater currents are themselves a magnetic risk.
3. **Magnetic hygiene is a whole-vehicle constraint.** R-10 applies to the AUV, not the payload. Thrusters, battery current loops and steel fasteners must be controlled at design time [13]; retrofitting a SERF payload to an existing vehicle is likely to fail on this basis alone.
4. **Shock and vibration unproven at system level.** The MEMS cell is robust [32, 4], but the optical bench, fibre alignment and coil formers must survive launch and recovery while holding the alignment tolerances of R-09.
5. **Short detection ranges against small targets.** At 22–68 m, area coverage rate is low and mission planning becomes demanding. For mine countermeasures this may be acceptable [14]; for wide-area search it may not, and that trade should be settled before investment.
6. **Heading dependence.** Any residual field couples through CAPE [6], so sensor gain may vary with vehicle heading, appearing operationally as a direction-dependent detection range. Heading error is a known issue in atomic magnetometry [8, 42] and demands in-situ calibration.
7. **Field calibration and drift.** Although the atomic transition is intrinsically calibrated [10], the surrounding chain — coil constants, axis alignment, compensation coefficients — is not. Compensation manoeuvres are needed to condition the Tolles–Lawson fit [17, 18], and these consume mission time.
8. **Validation data availability.** Compensation performance rests on a single published UUV benchmark [13]; independent replication on the target vehicle is needed before performance claims are made.
9. **Technology readiness.** The individual pieces are demonstrated [4, 13, 35]; an integrated, ruggedised, vehicle-mounted SERF gradiometer with active nulling is not. Treat as TRL 3–4, with integration risk concentrated in the nulling servo.

9.3 Conditions under which the concept fails

It is worth being explicit. The concept does not close if any of the following proves true: sustained 94 dB suppression cannot be maintained through vehicle manoeuvres; vehicle magnetic residual cannot be reduced to the few-nanotesla class on the target platform [13]; the thermal budget is incompatible with mission endurance; or operational requirements demand detection ranges against low-signature targets substantially beyond Table 5. Each is testable early and cheaply relative to full development, and the recommended programme structure is to test them first.

10 Recommendations and Next Steps

1. Extend the sensor model with atom number and photon shot noise [1, 40] to enable absolute sensitivity prediction and closure against published devices [4].
2. Implement closed-loop nulling with explicit servo dynamics [35, 7] and simulate loss of SERF lock under transient manoeuvres — the principal unmodelled failure mode.
3. Replace the parametric rejection ratio with a geometry-based vehicle magnetic model [15, 16] for the specific target platform.
4. Bench-validate CAPE using a shielded MEMS cell on a rotation stage; this directly tests Eq. (4), the report's central claim, at modest cost, following the experimental approach of [6].
5. Extend GPU validation beyond the single Ampere-class device verified here to the multi-GPU and non-Ampere configurations a deployed processing chain would plausibly use.
6. Obtain measured signature data for the small-target classes to replace the two engineering estimates.
7. Conduct an early magnetic survey of the candidate AUV [13] to establish whether R-10 is achievable before further investment.

11 Conclusion

A ruggedised SERF MEMS magnetometer is a physically credible AUV MAD payload. Its feasibility is governed not by sensor sensitivity, which exceeds requirement by five orders of magnitude [4, 13], but by ambient-field control and platform magnetic cleanliness. The nominal SERF field window [6] is shown to be insufficient by a factor of 5.3 once the Cross-Axis Projection Error is accounted for, giving a 94 dB suppression requirement that drives the majority of the component specification. All twenty-seven verification, validation and stress tests pass, including agreement between an independent quantum reference model [26] and the production solver [8] to 2.8×10^{-16}, and an emergent range-law exponent of 0.244 against a predicted 0.25 [19]. The barriers to implementation are identified and concentrated in the nulling servo and whole-vehicle magnetic design; each is testable early, and testing them first is the recommended programme structure.

References

[1] J. C. Allred, R. N. Lyman, T. W. Kornack, and M. V. Romalis, "High-sensitivity atomic magnetometer unaffected by spin-exchange relaxation," *Phys. Rev. Lett.* **89**(13), 130801 (2002). DOI:10.1103/PhysRevLett.89.130801.

[2] I. K. Kominis, T. W. Kornack, J. C. Allred, and M. V. Romalis, "A subfemtotesla multichannel atomic magnetometer," *Nature* **422**(6932), 596–599 (2003). DOI:10.1038/nature01484.

[3] H. B. Dang, A. C. Maloof, and M. V. Romalis, "Ultrahigh sensitivity magnetic field and magnetization measurements with an atomic magnetometer," *Appl. Phys. Lett.* **97**(15), 151110 (2010). DOI:10.1063/1.3491215.

[4] W. C. Griffith, S. Knappe, and J. Kitching, "Femtotesla atomic magnetometry in a microfabricated vapor cell," *Opt. Express* **18**(26), 27167–27172 (2010). DOI:10.1364/OE.18.027167.

[5] V. Shah, S. Knappe, P. D. D. Schwindt, and J. Kitching, "Subpicotesla atomic magnetometry with a microfabricated vapour cell," *Nature Photonics* **1**(11), 649–652 (2007). DOI:10.1038/nphoton.2007.201.

[6] A. Borna, J. Iivanainen, T. R. Carter, J. McKay, S. Taulu, J. Stephen, and P. D. D. Schwindt, "Cross-axis projection error in optically pumped magnetometers and its implication for magnetoencephalography systems," *NeuroImage* **247**, 118818 (2022). DOI:10.1016/j.neuroimage.2021.118818.

[7] A. Borna *et al.*, "Cross-axis dynamic field compensation of optically pumped magnetometer arrays for MEG," *NeuroImage* **262**, 119559 (2022). DOI:10.1016/j.neuroimage.2022.119559.

[8] S. J. Seltzer, *Developments in Alkali-Metal Atomic Magnetometry*, Ph.D. dissertation, Department of Physics, Princeton University (2008).

[9] I. M. Savukov and M. V. Romalis, "Effects of spin-exchange collisions in a high-density alkali-metal vapor in low magnetic fields," *Phys. Rev. A* **71**, 023405 (2005). DOI:10.1103/PhysRevA.71.023405.

[10] D. Budker and M. V. Romalis, "Optical magnetometry," *Nature Physics* **3**(4), 227–234 (2007). DOI:10.1038/nphys566.

[11] J. Li, W. Quan, B. Zhou, Z. Wang, J. Lu, Z. Hu, G. Liu, and J. Fang, "SERF atomic magnetometer – recent advances and applications: a review," *IEEE Sensors Journal* **18**(20), 8198–8207 (2018).

[12] W. Happer and A. C. Tam, "Effect of rapid spin exchange on the magnetic-resonance spectrum of alkali vapors," *Phys. Rev. A* **16**, 1877 (1977).

[13] M. Oehler *et al.*, "Processing of unmanned underwater vehicle vector magnetometer data," *Geophysical Prospecting* **72** (2024). DOI:10.1111/1365-2478.13556.

[14] M. Seidel *et al.*, "Underwater UXO detection using magnetometry on hovering AUVs," *Journal of Applied Geophysics* / BASTA project reporting (2023).

[15] W. E. Tolles and J. D. Lawson, "Magnetic compensation of MAD equipped aircraft," Airborne Instruments Laboratory, Rept. 201-1 (1950).

[16] A. R. Gnadt, A. B. Wollaber, and A. P. Nielsen, "Derivation and extensions of the Tolles-Lawson model for aeromagnetic compensation," arXiv:2212.09899 (2022).

[17] P. Leliak, "Identification and evaluation of magnetic-field sources of magnetic airborne detector equipped aircraft," *IRE Trans. Aerospace and Navigational Electronics* **8**(3), 95–105 (1961).

[18] B. W. Leach, "Aeromagnetic compensation as a linear regression problem," in *Information Linkage Between Applied Mathematics and Industry II*, Academic Press (1980).

[19] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, New York (1998), Ch. 5.

[20] W. M. Wynn *et al.*, "Advanced superconducting gradiometer/magnetometer arrays and a novel signal processing technique," *IEEE Trans. Magnetics* **11**(2), 701–707 (1975).

[21] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I*, Wiley (2001).

[22] S. M. Kay, *Fundamentals of Statistical Signal Processing, Vol. II: Detection Theory*, Prentice Hall (1998).

[23] National Centers for Environmental Information (NOAA), "Geomagnetism frequently asked questions," and the International Geomagnetic Reference Field (IGRF-13). Earth field intensity 25,000–65,000 nT.

[24] G. Lindblad, "On the generators of quantum dynamical semigroups," *Commun. Math. Phys.* **48**(2), 119–130 (1976).

[25] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002).

[26] J. R. Johansson, P. D. Nation, and F. Nori, "QuTiP 2: a Python framework for the dynamics of open quantum systems," *Comput. Phys. Commun.* **184**, 1234 (2013).

[27] J. C. Butcher, *Numerical Methods for Ordinary Differential Equations*, 3rd ed., Wiley (2016).

[28] J. Diebel, "Representing attitude: Euler angles, unit quaternions, and rotation vectors," Stanford University technical report (2006).

[29] G. Marsaglia, "Choosing a point from the surface of a sphere," *Ann. Math. Statist.* **43**(2), 645–646 (1972).

[30] W. L. Oberkampf and C. J. Roy, *Verification and Validation in Scientific Computing*, Cambridge University Press (2010).

[31] S.-K. Lee and M. V. Romalis, "Calculation of magnetic field noise from high-permeability magnetic shields and conducting objects with simple geometry," *J. Appl. Phys.* **103**(8), 084904 (2008).

[32] L.-A. Liew, S. Knappe, J. Moreland, H. Robinson, L. Hollberg, and J. Kitching, "Microfabricated alkali atom vapor cells," *Appl. Phys. Lett.* **84**(14), 2694–2696 (2004).

[33] P. D. D. Schwindt, S. Knappe, V. Shah, L. Hollberg, J. Kitching, L.-A. Liew, and J. Moreland, "Chip-scale atomic magnetometer," *Appl. Phys. Lett.* **85**(26), 6409–6411 (2004).

[34] S. J. Seltzer and M. V. Romalis, "Unshielded three-axis vector operation of a spin-exchange-relaxation-free atomic magnetometer," *Appl. Phys. Lett.* **85**(20), 4804–4806 (2004).

[35] J. Fang and J. Qin, "In situ triaxial magnetic field compensation for the spin-exchange-relaxation-free atomic magnetometer," *Rev. Sci. Instrum.* **83**(10), 103104 (2012).

[36] J. Fang, R. Li, L. Duan, Y. Chen, and W. Quan, "Study of the operation temperature in the spin-exchange relaxation free magnetometer," *Rev. Sci. Instrum.* **86**, 073116 (2015).

[37] V. Shah and M. V. Romalis, "Spin-exchange relaxation-free magnetometry using elliptically polarized light," *Phys. Rev. A* **80**, 013416 (2009).

[38] K. Wei, T. Zhao, X. Fang, *et al.*, "The suppression of the probe laser pumping effect in SERF atomic magnetometer," *Frontiers in Physics* **11**, 1181399 (2023).

[39] E. Boto *et al.*, "Moving magnetoencephalography towards real-world applications with a wearable system," *Nature* **555**(7698), 657–661 (2018).

[40] M. W. Mitchell and S. Palacios Alvarez, "Colloquium: Quantum limits to the energy resolution of magnetic field sensors," *Rev. Mod. Phys.* **92**, 021001 (2020).

[41] Y. Ito, D. Sato, K. Kamada, and T. Kobayashi, "Optimal densities of alkali metal atoms in an optically pumped K-Rb hybrid atomic magnetometer," *Opt. Express* **24**(14), 15391 (2016).

[42] E. B. Alexandrov *et al.*, "Laser pumping in the scheme of an Mx-magnetometer," *Optics and Spectroscopy* **78**, 292 (1995).

[43] T. R. Clem, "Superconducting magnetic sensors for mine detection and classification," Naval Surface Warfare Center, Coastal Systems Station, Panama City (1997).

[44] J. J. Holmes, *Exploitation of A Ship's Magnetic Field Signatures*, Synthesis Lectures on Computational Electromagnetics, Morgan & Claypool (2006).

[45] National Institute of Standards and Technology, "Microfabricated atomic sensors / microfabricated optically pumped magnetometers," NIST programme description (accessed 2026).